

RUTWIK JAIN

Curriculum Vitae

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Education

- 2021 **Ph.D. Computer Science**
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University of Wisconsin–Madison GPA – **3.82/4.0**
- Present
○ Ongoing Research: Power Excursion Management for GPUs
○ Advisors: Prof. Matt Sinclair and Prof. Shivaram Venkataraman
○ Coursework: Advanced Computer Architecture, HPC and Applications in Engineering, Computer Graphics, Operating Systems, Distributed Systems, HW/SW for Next Gen AI Systems, Fault-Tolerant Computing, Real-Time Computing Systems, Optimization, Data Visualization
- 2015 **B.E.(Hons.) Electrical and Electronics Engineering**
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Birla Institute of Technology and Science, Pilani GPA – **9.52/10.0**
- 2019
○ Class Rank: 3/93
○ Coursework: Computer Architecture, Embedded Systems Design, Digital Design, Digital Signal Processing, Microprocessors and Interfacing, Analog and Digital VLSI Design

Publications

In Conference Proceedings

- 2024 **Rutwik Jain**, Brandon Tran, Keting Chen, Matthew D. Sinclair, and Shivaram Venkataraman. PAL: A Variability-Aware Policy for Scheduling ML Workloads in GPU Clusters. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage, and Analysis*, SC '24. IEEE Press, 2024.
- 2022 Prasoon Sinha, Akhil Guliani, **Rutwik Jain**, Brandon Tran, Matthew D. Sinclair, and Shivaram Venkataraman. Not All GPUs Are Created Equal: Characterizing Variability in Large-Scale, Accelerator-Rich Systems. In *Proceedings of the International Conference on High Performance Computing, Networking, Storage and Analysis*, SC '22. IEEE Press, 2022.
- 2019 **Rutwik Jain** and Umang Garg. Modelling and Simulation of Onboard Wire Antennas for a 3U CubeSat (undergraduate). In *2019 Photonics and Electromagnetics Research Symposium - Fall (PIERS - Fall 2019)*, pages 871–879, Dec 2019.
- 2018 **Rutwik Jain**, Shubham Sharma, Kushagra Aggarwal, Tushar Goyal, Abhinav Sundhar, Ujjwal Anand, Nishad Sahu, and Joy Parikh. Modes of Operation for a 3U CubeSat with Hyperspectral Imaging Payload (undergraduate). In *Proceedings of the 69th International Astronautical Congress (IAC) 2018, Bremen, Germany*, Oct 2018.

Journal Articles

- 2023 Matthew D. Sinclair, Parthasarathy Ranganathan, Gaurang Upasani, Adrian Sampson, David Patterson, **Rutwik Jain**, Nidhi Parthasarathy, and Shaan Shah. Fifty Years of the International Symposium on Computer Architecture: A Data-Driven Retrospective. *IEEE Micro*, volume 43, pages 109–124. IEEE Computer Society, nov 2023.
- 2020 Anuj K. Ojha, **Rutwik Jain**, and Praveen Kumar AV. Magnetic quadrupole mode excitation of a cylindrical dielectric resonator antenna using planar feed. *Microwave and Optical Technology Letters*, volume 62, pages 484–490, 2020.

Work Experience

- May 2024 **Intern, AMD Research and Advanced Development (RAD)**
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Present Power Efficiency for AI Compute (PEAC) Team Austin, TX
- Worked with Srilatha (Bobbie) Manne's team to develop an event-based model for accurate on-chip power estimation on AMD Instinct GPUs.
 - Used the power estimation model to improve excursion management, particularly for LLM workloads.
 - Developed a compiler-based instruction injection mechanism to achieve frequency boosting by reducing power/activity transients during kernel launches.
- Jul 2019 **ASIC Design Engineer, NVIDIA**
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Jul 2021 Memory Subsystem Design Team Bangalore, India
- Microarchitecture design and RTL implementation for the SoC's memory management IP.
 - This included new feature implementation, modification of existing μ Arch for PPA optimization, logic synthesis for area and timing accuracy, iterative design to incorporate DV and PD feedback.
- Jan 2019 **ASIC Intern, NVIDIA**
↓
Jun 2019 Memory Subsystem Design Team Bangalore, India
- Developed a debug tool that uses a generated waveform database to identify, trace and correlate transactions as they flow through the Memory Management Unit (MMU).
 - Presented my work on developing and scaling the tool in a lightning talk at India-level internal NVIDIA conference, NTech 2019.

Research

- Aug 2021 **Characterizing Variability in Large-scale, Accelerator-rich Systems**
↓
Aug 2022 Advisors: Prof. Matt Sinclair and Prof. Shivaram Vankataraman
- Characterized GPU variability in GPU supercomputers such as ORNL Summit and analyzed how it is affected by cluster attributes like size, cooling, and GPU vendors.
 - Profiled different applications (image processing, machine translation, molecular dynamics, graph analytics) on the same cluster to evaluate application-dependence.
 - Co-authored and presented our work at Supercomputing (SC), 2022.
- Aug 2022 **PAL: Variability-aware Scheduling for Large-scale, GPU-based Systems**
↓
May 2024 Advisors: Prof. Matt Sinclair and Prof. Shivaram Vankataraman
- Designed a variability-aware scheduling policy that makes application-specific GPU allocations when scheduling jobs in large-scale GPU-based systems, based on variability profiles from hardware.
 - Authored and presented our work at Supercomputing (SC), 2024.
- May 2024 **Excursion Management in Accelerators**
↓
Present In collaboration with AMD Research
- Developed an accurate event-based on-chip power estimation methodology to detect power excursions on Instinct GPUs.
 - Obtained $< 13\%$ prediction errors across state-of-the-art deep learning workloads including LLaMA training, inference and diffusion models.
 - Used this digital power meter to demonstrate potential for effectively managing power excursions.

Course Projects

- Aug 2023 **Parallelized Implementation of Coordinated Animal Motion – Boid Flocking**
↓
Dec 2023 Course: CS 759 High Performance Computing and Applications in Engg, Fall 2023
- Wrote CPU-based C++ and OpenMP implementations and CUDA-based GPU implementation of flocking behavior to compare frame rate performance and scalability.
 - The CUDA kernel with additional shared memory optimizations allowed flocking simulations to scale gracefully as number of particles increases up to 2^{14} .
- Aug 2022 **PICLE: Prefetcher for Inter-CU Locality Exploitation in GPU TLBs**
↓
Dec 2022 Course: CS 757 Adv Computer Architecture II, Fall 2022
- Implemented a GPU L1-TLB prefetcher based on Valkyrie [Baruah et.al, PACT'20] which exploits inter-TLB locality to improve GPU virtual memory performance using gem5.

- Jan 2022 **Project Eagle: Open-source HW/SW Stack for AI Applications**
↓ Course: CS 838 Next-Gen HW/SW Design for AI/ML, Spring 2022
- May 2022 ○ Implemented RTL for a dual-ported, lockup-free L1 cache with prefetching for the memory system of UPCYCLE, a tiled accelerator architecture.
- Jan 2022 **Replicated Block Store**
↓ Course: CS 739 Distributed Systems, Spring 2022
- May 2022 ○ Implemented a primary-backup replica based distributed block store.
○ Analyzed and reported read and write performance, crash recovery and reintegration time, and evidence for correctness and linearizability guarantees.

Skills

Programming: C, C++, OpenMP | Scripting: Python | RTL Design: Verilog
GPU computing: CUDA, ROCm stack | HPC: MPI | Optimization: Julia & JuMP
Cycle-level Simulators: gem5 | RTL Design and Debug: ModelSim, VCS Synopsys
Version Control: Git | Containers: Docker, Apptainer | Visualization: Tableau

Talks & Presentations

- Jan 2025 **Characterizing GPU variability in Large-Scale, Accelerator-Rich Systems**, MLCommons Science and HPC Working Group, Online.
- Jan 2025 **PAL: A Variability-Aware Policy for Scheduling ML Workloads in GPU Clusters**, HPC, Systems and Applications Sync, AMD, Online.
- Dec 2024 **Excursion management in MI Accelerators**, AMD Research, Online.
- Nov 2024 **PAL: A Variability-Aware Policy for Scheduling ML Workloads in GPU Clusters**, SC 2024. Atlanta, GA.
- Jan 2024 **GPU Variability: Characterization Study and Mitigation Strategies**, AMD Research. Online.
- Nov 2022 **Not All GPUs are Created Equal: Characterizing GPU variability in Large-Scale, Accelerator-Rich Systems**, SC 2022. Dallas, TX.
- Oct 2022 **Variability-Aware Scheduling for Efficient Exascale Computing**, UW-Madison's Architecture Affiliates 2022. Madison, WI.
- Oct 2018 **Modes of Operation of a 3U CubeSat with Hyperspectral Imaging Payload**, 69th International Astronautical Congress (IAC) 2018. Bremen, Germany.

Teaching

- FA 21 **CS/ECE 552: Introduction to Computer Architecture**
- SP 22 Taught Verilog and HDL concepts in discussion section. Also helped students with in-class exercises and debugging projects (pipelined processor and cache implementation)
- FA 22 **CS/ECE 252: Introduction to Computer Engineering**
Assisted students with homework and projects through office hours, graded homework assignments.

Awards & Honors

- Received CS Summer Research Assistantship from UW-Madison for the 2022 summer term.
- First-year CS Departmental Scholarship from UW-Madison for academic year 2021-22.
- Merit Scholarship at BITS Pilani: 40% tuition waiver for seven semesters from 2015 to 2019.